TWO POINTS OF A SPECTRAL SPACE EITHER HAVE A COMMON SPECIALIZATION OR DISJOINT NEIGHBOURHOODS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the constructible topology of a spectral space is Hausdorff and quasi-compact. Invoked inside the counted proof (source lines 4731-4733). Supplies the compactness used to intersect the neighbourhood families.
- The statement that an intersection of closed subsets of a quasi-compact space is nonempty when all finite intersections are. Invoked inside the counted proof (source lines 4733-4734). Justifies the nonemptiness of the total intersection.

Source reference. Original lemma lemma-constructible-hausdorff-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4522–4522. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-hausdorff-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-intersection-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1941–1941. Navigation pointer only: the counted text refers to the Stacks item lemma-intersection-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Lemma 0.1. *Let X be a spectral space. Let $x, y \in X$. Then either there exists a third point specializing to both x and y, or there exist disjoint open neighbourhoods containing x and y.*

Proof. Let $\{U_i\}$ be the set of quasi-compact open neighbourhoods of x. A finite intersection of the U_i is another one. Let $\{V_j\}$ be the set of quasi-compact open neighbourhoods of y. A finite intersection of the V_j is another one. If $U_i \cap V_j$ is empty for some i, j we are done. If not, then the intersection $U_i \cap V_j$ is nonempty for all i and j. The sets $U_i \cap V_j$ are closed in the constructible topology on X. By Lemma lemma-constructible-hausdorff-quasi-compact we see that $\bigcap(U_i \cap V_j)$ is nonempty (Lemma lemma-intersection-closed-in-quasi-compact). Since X is a sober space and $\{U_i\}$ is a fundamental system of open neighbourhoods of x, we see that $\bigcap U_i$ is the set of generalizations of x. Similarly, $\bigcap V_j$ is the set of generalizations of y. Thus any element of $\bigcap(U_i \cap V_j)$ specializes to both x and y. $\square$